# Final Course Project Guidelines

CS4082: Machine Learning

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Effat University · Computer Science Department · Spring 2026

## Learning Objectives

The goal of the final course project is to:

- Apply machine learning techniques to solve real-world problems in healthcare domains
- Develop expertise in data preprocessing, feature engineering, model development, evaluation, and interpretability
- Foster research-based thinking with potential for academic publication
- Enhance teamwork, critical problem-solving, and professional technical reporting skills
- Align with Saudi Vision 2030 goals in digital transformation and AI innovation

## 1 Team Formation

- Group members: **3–4 students**
- **Equal contribution** from each member is mandatory and must be documented
- Each member must clearly state their individual contributions in the final report

## 2 Project Deliverables

### 1. Code Repository (GitHub)

- Well-organized directory structure
- Comprehensive `README.md` with: project description, installation instructions, usage examples, dependencies list
- Clean, documented code
- Jupyter notebooks with explanations
- Requirements file (`requirements.txt`)

### 2. Technical Report (Max 10 pages)

**Format:** ACL Submission Template
**Template:** Overleaf ACL Template

**Required Sections:** Abstract (150–250 words), Introduction & Motivation, Related Work, Data Description & Preprocessing, Methodology, Experimental Results & Discussion, Conclusion & Future Work, References

### 3. Presentation & Poster

**Oral Presentation:** 12–15 minutes + Q&A, all members must present, professional slide design, live demo encouraged

**Research Poster:** A1 or A0 format, visual representation of work, key findings highlighted, QR code linking to repository

### 4. Portfolio Website

Project overview, interactive visualizations, key findings, team profiles, and links to GitHub repository, technical report (PDF), and demo video (optional). Suggested platforms: GitHub Pages, Streamlit, or Hugging Face Spaces.

## 3 Project Topics

### Meta-Project Theme

**Learning Under Adversarial and Non-Stationary Data**

**Core Research Question:**

*How do learning models remain robust when data distributions evolve, degrade, or are manipulated?*

**Important:** "Adversarial" does **NOT** mean "Hacker." In machine learning, adversarial conditions refer to challenges that compromise model reliability:

**Concept Drift** • **Label Noise** • **Data Poisoning** • **Distribution Shift** • **Minority Class Suppression**

### Healthcare Topics Overview

### Expected ML Techniques Across All Projects

**Classification Algorithms:**

- Logistic Regression
- Decision Trees / Random Forest
- Support Vector Machines (SVM)
- K-Nearest Neighbors (KNN)
- Naive Bayes
- Ensemble Methods (Bagging, Boosting)

**Key Techniques to Apply:**

- Train/Test Split, Cross-Validation
- Feature Scaling & Selection
- SMOTE & Class Weighting
- Hyperparameter Tuning (Grid Search)
- Evaluation Metrics (Precision, Recall, F1, AUC)
- Visualization (Confusion Matrix, ROC Curves)

| # | Topic | ML Paradigm | Adversarial Challenge |
|---|---|---|---|
| 1 | Clinical Outcome Prediction from Noisy Medical Records | Supervised Classification | Label Noise |
| 2 | Health Risk Prediction Across Heterogeneous Patient Populations | Supervised Classification | Distribution Shift |
| 3 | Early Detection of Rare Medical Conditions | Supervised Classification | Minority Class Suppression |
| 4 | Patient Condition Monitoring under Evolving Data Patterns | Classification + Drift Detection | Concept Drift |
| 5 | Automated Medical Diagnosis with Corrupted or Manipulated Training Data | Supervised Classification | Data Poisoning |
| 6 | Robust Health Status Prediction under Multiple Adversarial Data Conditions | Ensemble Learning | Combined Challenges |

Table 1: Disease-Agnostic Healthcare Machine Learning Project Topics

| Criterion | Weight | Description |
|---|---|---|
| Technical Implementation | 25% | Code quality, algorithm implementation, proper use of ML techniques, reproducibility |
| Methodology & Rigor | 20% | Sound experimental design, appropriate preprocessing, model selection justification |
| Results & Analysis | 20% | Comprehensive evaluation metrics, insightful analysis, honest discussion of limitations |
| Technical Report | 15% | Writing quality, proper structure, adherence to ACL format, citations |
| Presentation & Poster | 10% | Clarity, visual design, ability to explain and answer questions |
| Portfolio Website | 5% | Professional design, clear documentation, accessibility |
| Innovation & Creativity | 5% | Novel approaches, creative solutions, going beyond baseline requirements |

Table 2: Project Evaluation Rubric (Total: 100%)

# 4 Evaluation Criteria

## Detailed Grading Rubric

| Criterion | Excellent (90-100%) | Good (75-89%) | Satisfactory (60-74%) | Needs Work (<60%) |
|---|---|---|---|---|
| Technical Implementation | Flawless code, advanced techniques, fully reproducible | Clean code, correct implementation, minor issues | Working code with some errors, basic techniques | Non-functional or poorly written code |
| Methodology | Rigorous design, justified choices, thorough preprocessing | Sound methodology, good preprocessing | Basic methodology, minimal preprocessing | Flawed or missing methodology |
| Results | Comprehensive metrics, deep insights, honest limitations | Good analysis, appropriate metrics | Basic results, limited analysis | Incomplete or incorrect results |
| Report | Publication-ready, excellent writing, perfect format | Well-written, minor formatting issues | Adequate writing, some structure issues | Poor writing, wrong format |
| Presentation | Engaging, clear, handles Q&A expertly | Good delivery, answers most questions | Adequate presentation, struggles with Q&A | Unclear, unprepared |

Table 3: Detailed Performance Rubric

## 5 Suggested Project Timeline

| Period | Milestone |
|---|---|
| Week 1–2 | Team Formation & Topic Selection |
| Week 3–4 | Literature Review & Data Collection |
| Week 5–6 | Data Preprocessing & EDA |
| Week 7–9 | Model Development & Training |
| Week 10–11 | Evaluation & Documentation |
| Week 12 | Final Submission |

## 6 Submission Guidelines

**File Naming:** `CS4082_Project_GroupX_TopicY.zip`

**Required Contents:**

- Technical Report (PDF, ACL format)
- Presentation slides (PDF/PPTX)
- Poster (PDF, A1/A0)
- GitHub repository link (in README)
- Portfolio website URL (in README)

## 7 Academic Integrity

- All submitted work must be **original**
- Properly cite all external sources, code libraries, and datasets
- AI tools (ChatGPT, Claude, Copilot) may be used for **learning assistance only**
- Plagiarism or unauthorized collaboration will result in **zero grade**
- Each member must sign a contribution declaration form

## 8 Support & Resources

**Contact:**

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**Recommended Resources:**

- Scikit-learn Documentation
- PyTorch/TensorFlow Tutorials
- Papers With Code
- Kaggle Datasets & Notebooks

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*"The goal is to turn data into information, and information into insight."*
— Carly Fiorina